

AI INFRASTRUCTURE

FULL EQUITY RESEARCH REPORT

**The AI CapEx Cycle Is Entering Its
Cash-Return Phase**

**AI Infrastructure Economics, Depreciation Timing and Return on Invested
Capital**

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Independent research on AI infrastructure, capital intensity, cash returns and valuation.

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1. Executive Summary

The AI infrastructure cycle is entering a more demanding stage.

Over the past several years, hyperscalers have been rewarded for accelerating capital expenditure, securing advanced chips and expanding data-center capacity. The next phase will be defined less by how much capital is deployed and more by whether those assets can generate sufficient utilization, revenue and cash returns.

The financial impact of this investment cycle arrives in stages. Capital expenditure reduces free cash flow when cash is spent, while depreciation reaches the income statement only after infrastructure is installed, tested and placed into service.

This creates a material timing gap:

CapEx is the upfront cash-flow burden, while depreciation is the delayed earnings burden.

Specialized cloud providers have indicated that contracted CPU or GPU deployments can sometimes recover equipment costs in approximately two years. However, these claims usually refer to hardware-level revenue pay-back under high utilization and long-term customer commitments.

They generally do not include the full cost of:

- data-center buildings;
- power infrastructure;
- cooling;
- networking;
- financing;
- maintenance;
- idle capacity;
- future equipment replacement.

The central investment question is therefore not simply whether AI-related revenue is growing.

It is whether AI-related revenue and cash contribution can grow faster than depreciation, infrastructure operating costs and replacement investment.

Our framework suggests that the main risk is not that every CPU or GPU investment fails to recover its cost. The larger risk is that the overall cost base compounds faster than the economic returns generated by the infrastructure.

2. Investment Thesis

The AI infrastructure cycle is transitioning from a buildout story into a return-on-capital story.

During the initial phase, investors focused primarily on:

- access to GPUs;
- data-center capacity;
- cloud revenue growth;
- AI product adoption;
- capital-spending commitments.

As infrastructure enters service, the focus should move toward:

- utilization;
- revenue generated per unit of deployed capital;
- incremental depreciation;
- cloud margins;
- free-cash-flow conversion;
- economic asset life;
- replacement CapEx;
- return on invested capital.

A hyperscaler can report strong AI revenue growth while still destroying shareholder value if the incremental revenue requires disproportionately high capital investment.

The long-term test is therefore:

Incremental AI cash contribution > operating costs + financing costs + maintenance CapEx + replacement CapEx

At the earnings level:

Incremental AI gross profit > incremental depreciation + infrastructure operating expenses

At the capital-return level:

AI ROIC > weighted average cost of capital

3. The Accounting Impact Arrives After the Cash Out-flow

AI infrastructure is made up of several asset categories with different accounting lives and economic characteristics.

3.1 Shorter-Lived Technology Equipment

This category includes:

- CPU servers;
- GPU and AI accelerators;
- HBM;
- DRAM;
- SSD and NVMe storage;
- network switches;
- optical modules;
- high-speed interconnects;
- server racks and related equipment.

CPU, GPU, HBM, DRAM and SSD are generally not depreciated separately in hyperscaler financial statements. They are typically capitalized as part of server or network equipment.

Large technology companies commonly depreciate server and network assets over approximately five to six years.

3.2 Medium-Life Infrastructure

This category includes:

- electrical distribution systems;
- cooling systems;
- transformers;
- backup generators;
- UPS systems;
- mechanical equipment;
- data-center networking infrastructure.

These assets may be depreciated over roughly 10 to 25 years.

3.3 Long-Life Infrastructure

This category includes:

- data-center buildings;
- permanent site improvements;
- campus infrastructure;
- land.

Buildings may be depreciated over 20 to 40 years, while land is generally not depreciated.

A single AI data center therefore contains several asset classes with different depreciation schedules.

The risk is not simply the accounting life of the asset. It is whether the asset retains sufficient economic value throughout that accounting life.

4. The AI Infrastructure Cycle Has Three Financial Phases

The economics of AI infrastructure can be viewed through three overlapping phases.

Phase	Balance-sheet and cash-flow impact	Earnings impact
Build	CapEx rises; cash declines; CIP and assets not yet in service increase	Limited initial depreciation
Deploy	CIP transfers into operating PP&E; infrastructure begins operating	Depreciation and operating expenses
Harvest or reinvest	Historical assets reach full-year depreciation while new capacity may still be added	Returns must absorb depreciation

4.1 Phase One: Build

During the build phase, companies:

- sign semiconductor and equipment contracts;
- purchase servers;
- build data centers;
- secure power capacity;
- install cooling and networking infrastructure;
- reserve future compute capacity.

The immediate financial effects include:

- higher investing cash outflows;
- lower free cash flow;
- rising construction in progress;
- rising assets not yet in service;
- relatively limited depreciation.

Cash may therefore leave the business well before the infrastructure has a meaningful impact on reported earnings.

4.2 Phase Two: Deploy

Depreciation generally begins when an asset reaches its intended operating condition and is placed into service.

A GPU system may have been delivered and paid for but still require:

- rack installation;
- electrical connection;
- liquid cooling;
- network configuration;
- optical interconnects;
- software deployment;
- testing and commissioning.

Once those steps are complete, the asset is transferred into operating PP&E and begins depreciating.

At that point:

- CIP may decline or grow more slowly;
- gross placed-in-service PP&E increases;
- depreciation enters cost of revenue or operating expenses;
- cloud and AI infrastructure margins may begin to compress.

The first year usually reflects only a partial period of depreciation.

If an asset enters service on April 1 and carries annual depreciation of \$1.2 billion, only nine months, or approximately \$0.9 billion, would be recognized in the first year.

The following year would include the full \$1.2 billion.

4.3 Phase Three: Harvest—or Reinvest

The third phase begins when multiple years of capital expenditure overlap in the depreciation schedule.

Assume servers are depreciated over five years:

Investment year	New server CapEx	Full-year depreciation
2026	\$50B	\$10B
2027	\$80B	\$16B
2028	\$100B	\$20B
2029	\$90B	\$18B

Even though CapEx falls from \$100 billion to \$90 billion in 2029, depreciation may continue increasing because the company is still depreciating assets deployed in prior years.

Under this simplified example:

$$| \quad \text{2029 depreciation} = \$10\text{B} + \$16\text{B} + \$20\text{B} + \$18\text{B} = \$64\text{B}$$

This illustrates an important investment point:

$$| \quad \text{CapEx growth can slow while depreciation growth continues to accelerate.}$$

A slowdown in capital spending therefore does not necessarily mean that margin pressure has peaked.

5. Understanding CIP, Assets Not Yet in Service and PP&E

5.1 Construction in Progress

Construction in progress, or CIP, represents long-lived assets that remain under construction, installation or testing.

Examples include:

- unfinished data centers;
- servers being installed;
- power infrastructure under construction;
- cooling systems being commissioned;
- optical networks being deployed.

CIP is included in PP&E on the balance sheet but generally does not depreciate before it reaches its intended use.

5.2 Assets Not Yet in Service

Assets not yet in service is a broader category that may include:

- CIP;
- completed facilities awaiting acceptance;
- delivered servers awaiting installation;
- installed equipment still under testing;
- infrastructure waiting for grid or network connection.

CIP is therefore generally a subset of assets not yet in service, although individual company definitions vary.

5.3 Gross Placed-in-Service PP&E

Gross placed-in-service PP&E represents the original capitalized cost of assets that have entered service, before deducting accumulated depreciation.

Rapid growth in gross operating PP&E generally signals that:

- new servers are coming online;
- new data-center phases are entering service;
- CIP is being transferred into operating asset categories;
- the depreciation base is expanding.

5.4 PP&E Depreciation

PP&E depreciation represents the periodic accounting expense assigned to operating fixed assets.

It is important to distinguish:

- current-period depreciation expense;
- accumulated depreciation.

Current-period depreciation appears in the income statement and cash-flow statement.

Accumulated depreciation appears on the balance sheet and represents all depreciation recorded since the assets entered service.

6. The Depreciation Cycle Lags the CapEx Cycle

The time between capital spending and depreciation depends on the asset category.

Asset category	Estimated CapEx-to-depreciation timing	Estimated timing to full-year depreciation
Servers added to an existing facility	1–6 months	7–18 months
Server, storage and network expansion	3–12 months	9–24 months
New data-center phase	12–24 months	18–36 months
Large campus and power infrastructure	24–48 months	30–60 months

These are analytical estimates rather than standardized corporate disclosures.

Actual timing depends on:

- chip availability;
- construction progress;
- permitting;
- electrical connections;
- cooling capacity;
- network readiness;
- testing;
- customer deployment.

Investors can assess the stage of the cycle using the following indicators:

Indicator	Interpretation
CapEx growth	Current spending intensity
CIP growth	Unfinished or uncommissioned project pipeline
Assets not yet in service	Future depreciation pipeline
Gross placed-in-service PP&E growth	Speed of infrastructure deployment
Depreciation growth	Rate at which prior CapEx is entering earnings
CapEx/depreciation	Current investment relative to existing asset consumption
Revenue/depreciation	Revenue productivity of the asset base

A high level of CIP combined with rapidly rising placed-in-service PP&E suggests that a company is simultaneously funding the next capacity wave and absorbing depreciation from the prior one.

This is the stage at which an infrastructure cycle begins to become an earnings-cycle issue.

7. Hardware Can Pay Back Quickly—but the System Takes Longer

There are credible industry cases in which CPU or GPU deployments achieve approximately two-year payback periods.

However, “two-year payback” may refer to:

- revenue payback;
- cash payback;
- equipment payback;
- fully loaded infrastructure payback.

These are not equivalent.

7.1 Revenue Payback

Revenue payback is typically calculated as:

$$\text{Hardware and deployment CapEx} \div \text{annual contracted revenue}$$

For example, a GPU cluster costing \$100 million and generating \$50 million of annual contracted revenue would have a two-year revenue payback.

This outcome is most plausible when infrastructure benefits from:

- high utilization;
- long-term contracts;
- take-or-pay commitments;
- stable pricing;
- available power;
- low financing costs.

7.2 Cash Payback

Cash payback is more demanding:

$$\text{Total deployed investment} \div \text{annual operating cash contribution}$$

Operating cash contribution deducts cash costs such as:

- electricity;
- cooling;
- networking;
- maintenance;
- personnel;
- facility expenses;
- financing costs.

A two-year revenue payback may therefore translate into a three-year or longer cash payback.

8. CPU Server Economics

Large cloud providers do not simply rent a CPU. They sell an integrated service that includes:

- virtual compute;
- memory;
- storage;
- networking;
- databases;
- security;
- availability;
- management services.

The relevant economic unit is a server node or rack.

Assume a CPU server has:

- deployed cost: \$40,000;
- annual revenue at expected utilization: \$50,000;
- infrastructure cash contribution margin: 50%.

Annual cash contribution would be approximately:

$$| \quad \$50,000 \times 50\% = \$25,000$$

Estimated payback:

$$| \quad \$40,000 \div \$25,000 = 1.6 \text{ years}$$

This demonstrates why a two-to-three-year equipment payback is plausible for highly utilized CPU servers operated by a scaled cloud provider.

However, lower utilization, customer discounts and higher energy costs can materially extend the return period.

9. GPU Server Economics

GPU systems can produce higher revenue per deployment but carry greater capital and operational intensity.

They require:

- advanced accelerators;
- HBM;
- high-speed networking;
- optical interconnects;
- high power density;
- sophisticated cooling;
- cluster management software.

When demand is contracted and supply is scarce, GPU hardware may achieve rapid revenue recovery.

However, GPU economics are more exposed to:

- customer concentration;
- utilization volatility;
- rental-price declines;
- rapid product cycles;
- energy costs;
- technological obsolescence.

The economic outcome is therefore more variable than the headline revenue opportunity may suggest.

10. Equipment Payback Does Not Equal Data-Center Payback

Even when CPU or GPU hardware recovers its cost quickly, the full infrastructure system may take substantially longer.

Investment layer	Illustrative payback range
Highly utilized CPU hardware	1.5–3 years
Highly utilized GPU hardware	Approximately 2–3 years
Servers, network and direct deployment	2–4 years
GPU capacity plus supporting facility infrastructure	3–4 years or longer
Fully developed data-center campus	4–8 years or longer

The full data-center investment includes:

- land;
- buildings;
- electrical infrastructure;
- transformers;
- cooling;
- backup power;
- networking;
- redundant capacity;
- security;
- long-term maintenance.

The appropriate conclusion is therefore:

A GPU can recover its hardware cost in approximately two years without the complete AI infrastructure investment recovering in two years.

11. Cash Payback and Accounting Life Are Different

An asset can achieve economic payback before it is fully depreciated.

Assume a server costs \$40,000 and is depreciated over five years:

Annual depreciation = \$8,000

If it produces \$25,000 in annual cash contribution:

Cash payback = 1.6 years

After two years, the business may have recovered the original investment, but the asset still carries a book value of:

\$40,000 – \$16,000 accumulated depreciation = \$24,000

The asset continues to generate revenue and cash flow while the remaining cost is recognized over the next three years.

This is economically attractive when the asset remains productive.

The risk emerges when its revenue-generating capability declines faster than its book value.

12. Utilization Is More Important Than Installed Capacity

Installed compute capacity does not automatically generate value.

Infrastructure creates value only when it is:

- billed to customers;
- used internally to produce measurable incremental revenue;
- used to improve strategically important existing businesses.

CPU infrastructure typically supports a broader set of workloads and can often be reassigned to less demanding applications.

GPU workloads may be more specialized and customer demand may be more concentrated.

Business models also matter.

A specialist AI cloud provider may deploy GPUs against a long-term customer contract, making return attribution relatively direct.

A hyperscaler may use AI infrastructure across:

- cloud services;
- model training;
- search;
- advertising;
- recommendation systems;
- productivity applications;
- internal research.

In these cases, returns may appear as:

- stronger customer retention;
- improved advertising conversion;
- higher product engagement;
- competitive protection.

The absence of separately reported AI revenue therefore makes return measurement more difficult.

13. Depreciation Is Only One Component of Economic Cost

The broader cost base includes:

- power;
- cooling;
- networking;
- maintenance;
- personnel;
- software;
- real estate;
- leases;
- financing;
- maintenance CapEx;
- replacement CapEx.

A server may recover its initial hardware cost quickly and still produce an unattractive overall return after the full cost stack is included.

This is especially important because the infrastructure cycle is continuous.

Hyperscalers must:

- maintain current capacity;
- replace aging hardware;
- deploy new architectures;
- upgrade power and cooling;
- secure additional data-center capacity.

The relevant framework is not:

AI revenue > hardware cost

It is:

AI cash contribution > operating costs + financing costs + maintenance CapEx + replacement CapEx

14. Economic Obsolescence May Arrive Before Depreciation Ends

A server may remain physically operational for five or six years while its economic value declines sooner.

New GPU generations can improve:

- performance;
- energy efficiency;
- memory capacity;
- interconnect bandwidth;
- cost per unit of compute.

Older equipment may then experience:

- lower rental pricing;
- reduced utilization;
- migration to lower-value workloads;
- shortened useful lives;
- accelerated depreciation;
- impairment or disposal losses.

The key risk is:

Economic asset life may be shorter than accounting asset life.

This is particularly relevant for AI accelerators because technology cycles are rapid and electricity efficiency has become increasingly important.

The next hardware cycle may begin before the prior generation has been fully depreciated.

A company may simultaneously absorb:

- depreciation on older equipment;
- CapEx for new equipment;
- lower utilization on legacy systems;
- ramp-up inefficiencies on new systems;
- higher power and cooling costs.

15. Principal Risks

15.1 Utilization Risk

If capacity is built ahead of demand, assets may begin depreciating before they generate sufficient revenue.

This creates a direct mismatch:

Depreciation and operating expenses rise faster than AI revenue and cash contribution.

15.2 Pricing Risk

Compute prices may decline as:

- supply expands;
- competition increases;
- proprietary accelerators gain adoption;
- customers gain bargaining power;
- inference efficiency improves.

If prices fall faster than hardware costs, payback periods extend and ROIC declines.

15.3 Revenue Attribution Risk

Some AI spending may protect existing businesses rather than generate standalone revenue.

AI infrastructure may be necessary to maintain:

- search quality;
- advertising performance;
- customer engagement;
- product competitiveness.

This spending may be strategically necessary but difficult to monetize directly.

15.4 Replacement-Cycle Risk

A new investment cycle may begin before the previous cycle has generated an adequate return.

This can turn AI infrastructure into a structurally high-capital-intensity business.

15.5 Financing Risk

Debt, leases, equipment financing and long-term power commitments can reduce equity returns.

Positive operating income does not necessarily translate into attractive shareholder returns after financing costs.

15.6 Full-System Payback Risk

A hardware deployment may perform well while the wider data-center facility remains underutilized.

Unused land, power and cooling capacity can depress returns for years.

16. The Most Concerning Financial Combination

Indicator	Risk signal
CapEx	Remains elevated
CIP	Continues growing
Placed-in-service PP&E	Accelerates
Depreciation	Rises rapidly
AI or cloud revenue	Growth slows
Cloud margin	Declines
Free cash flow	Remains constrained
Compute utilization	Weakens
Compute pricing	Declines
Financing costs	Increase

This combination would indicate:

The company is simultaneously funding the next infrastructure wave and absorbing the earnings burden of the previous one.

17. Key Monitoring Indicators

Metric	Healthy signal	Risk signal
CapEx growth	Moderates over time	Remains structurally elevated
CIP	Stabilizes or falls	Continues rapid expansion
Depreciation growth	Below revenue growth	Above revenue growth
Cloud margin	Stable or expanding	Persistently declining
Free cash flow	Recovers	Remains constrained
Revenue/depreciation	Improves	Deteriorates
CapEx/revenue	Declines	Continues rising
Incremental AI gross profit/new depreciation	Above 1 and improving	Below 1 or declining
Useful-life changes	Stable	Shortened useful lives or impairments
Utilization	High and stable	Weakening or volatile

18. Valuation Implications

The AI buildout can influence valuation through several channels.

18.1 Lower Near-Term Free Cash Flow

Elevated capital spending reduces free cash flow even if reported operating income remains strong.

This can reduce:

- FCF yield;
- share-repurchase capacity;
- balance-sheet flexibility;
- near-term valuation support.

18.2 Lower Future Margins

As assets enter service, rising depreciation and operating costs may pressure:

- gross margin;
- cloud margin;
- operating margin;
- EPS growth.

18.3 Higher Reinvestment Requirements

If AI infrastructure requires frequent replacement, normalized CapEx may remain structurally higher than historical levels.

This would reduce sustainable free cash flow relative to accounting earnings.

18.4 Lower Terminal Returns

If returns on AI infrastructure approach or fall below the cost of capital, higher revenue growth may not justify a higher valuation multiple.

Investors should therefore focus less on reported EBITDA and more on:

- normalized free cash flow;
- maintenance and replacement CapEx;
- incremental ROIC;
- revenue productivity of the asset base.

19. Bull, Base and Bear Cases

19.1 Bull Case

The positive scenario requires:

- high utilization;
- stable compute pricing;
- strong AI demand;
- revenue growth above depreciation growth;
- improving cloud margins;
- moderating CapEx;
- recovering free cash flow;
- equipment payback within two to three years.

Under this scenario, AI infrastructure becomes a high-return asset base that continues generating cash after initial payback.

19.2 Base Case

The base case assumes:

- demand remains healthy;
- utilization remains adequate;
- depreciation rises;
- CapEx growth moderates only gradually;
- cloud margins remain under pressure;
- free cash flow improves slowly.

AI investment creates value, but the return period is longer and capital intensity remains elevated.

19.3 Bear Case

The negative scenario includes:

- overcapacity;
- weaker utilization;
- falling compute prices;
- rapid technological replacement;
- higher depreciation;
- elevated CapEx;
- weaker free cash flow;
- declining cloud margins;
- asset-life reductions or impairments.

Under this scenario, the AI buildout creates strong revenue growth but weak shareholder returns.

20. Conclusion

The AI infrastructure cycle should be understood as a long-duration capital process:

| Contracting → cash payment → CapEx → CIP or assets not yet in service → installation and testing → placed-in-service PP&E → depreciation → revenue and cash generation → replacement or retirement

CPU and GPU equipment can plausibly recover their initial costs in approximately two to three years under high utilization, stable pricing and contracted demand.

However, equipment-level payback does not demonstrate that the full data-center system earns an attractive return.

The central questions are:

- How quickly is capacity entering service?
- Is utilization sufficiently high?
- Can compute pricing remain resilient?
- Is incremental revenue growing faster than depreciation?
- Does cash contribution cover power, maintenance and financing?
- Will replacement CapEx absorb a large share of future returns?
- Is economic asset life shorter than accounting life?

The market's focus is likely to shift from:

| AI Revenue Growth

toward:

| AI Utilization, Cash Returns and Return on Invested Capital.